

GATE ES 2024 PAPER COMPLETE SOLUTION

NOTE:

A Special thanks to KP Sir for providing the solutions of all the questions

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Our Materials are Print-Friendly

GOOD LUCK FOR THE 2025 EXAMINATION!!!!!!!!!!

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General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

English vocab Q.

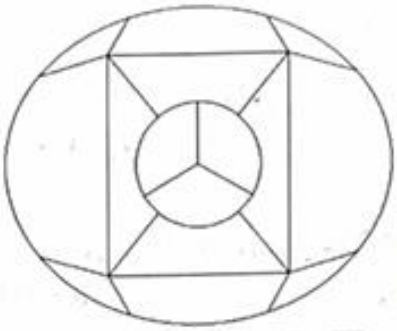
Q.1	If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft]. Which one of the given options is appropriate to fill the blank?
~~(A)~~	frown → Negative Expression (Sadness, worry, anger)
~~(B)~~	fawn → To show affection
~~(C)~~	vein → Blood vessel
~~(D)~~ ✓	vain → useless / unsuccessful

Sick → infirm → moribund
(Bimaar) (weak) (near death)

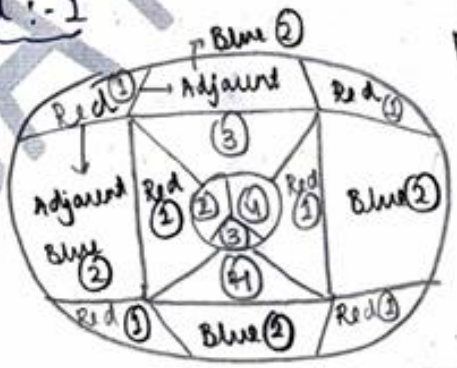
Silly → vain → daft
(Nadaan) (foolish)

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Read the question properly

Q.2	The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is 
~~(A)~~	4
(B)	3
(C)	5
(D)	6

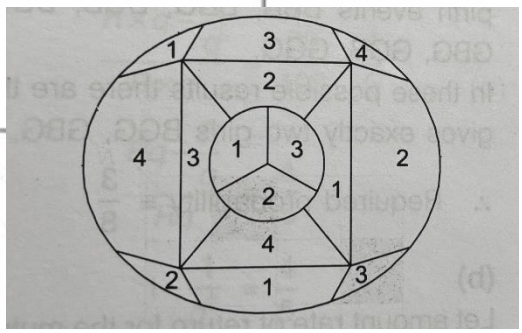
Method :- 1



Excluding ✱ ✱ Corners

only take side ways Adjacent

4 colours



Method:2 Four Colour Theorem

No more than 4 colours are required to colour the regions of any map so that no two adjacent region have same colour. → [A/B] ✓ [C & D] ✗

Permutations & Combinations

Q.3	How many 4-digit positive integers divisible by 3 can be formed using only the digits {1, 3, 4, 6, 7}, such that no digit appears more than once in a number?
(A)	24
(B)	48
(C)	72
(D)	12

Soln:

① Given → 5 digit

Req → 4 digit w/o repetition which % by 3 (divisible)

② {1, 3, 4, 6, 7}

case 1: Remove 7 ⇒ {1, 3, 4, 6} = Sum ⇒ Σ = 14 (Not divisible by 3) ✗

case 2: Remove 6 ⇒ {1, 3, 4, 7} : Sum ⇒ Σ = 15 (Divisible by 3) ✓

case 3: Remove 4 ⇒ {1, 3, 6, 7) = Sum ⇒ Σ = 17 (Divisible by 3) ✗

case 4: Remove 3 ⇒ {1, 4, 6, 7) = Sum ⇒ Σ = 18 (Divisible by 3) ✓

case 5: Remove 1 ⇒ {3, 4, 6, 7) = Sum ⇒ Σ = 20 (Divisible by 3) ✗

③ {1, 3, 4, 7} ⇒ 4! = 4 × 3 × 2 × 1 = 24

{1, 4, 6, 7) ⇒ 4! = 4 × 3 × 2 × 1 = 24

Ans ⇒ 24 + 24 = 48

Infinite Series

Q.4	The sum of the following infinite series is $2+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{8}+\frac{1}{9}+\frac{1}{16}+\frac{1}{27}+\cdots$
(A)	11/3
~~(B)~~	7/2
(C)	13/4
(D)	9/2

Soln : $2 + 1/2 + 1/3 + 1/4 + 1/8 + 1/9 + 1/16 + 1/27 + \cdots$

$= 2 + [1/2 + 1/4 + 1/8 + 1/16 + \cdots] + [1/3 + 1/9 + 1/27 + \cdots]$

Summation of infinite GP series formula: $S_\infty = \frac{a}{1-r}$ $\qquad S_\infty = \frac{a}{1-r}$

$= \left[\frac{1/2}{1-1/2}\right] \qquad = \left[\frac{1/3}{1-1/3}\right]$

Ans $= 2 + \left[\frac{1/2}{1-1/2}\right] + \left[\frac{1/3}{1-1/3}\right]$

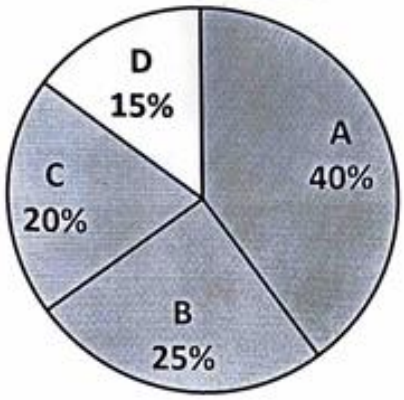
$= 2 + 1 + 1/2$

$= 7/2$

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Data Interpretation

Q.5 In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid.



Based on the data provided, the total number of valid votes received by the candidates B and C is

(A)	45,000
(B)	49,500
(C)	51,750
(D)	54,000

Soln :-

Total votes = 1,15,000
Invalid votes = 5,000
valid votes = 1,10,000

Share of valid votes
A = 40%.
B = 25%. } 45% ≃ 49,500 * Ans
C = 20%.
D = 15%.

valid votes from B & C
= 49,500

Q.6 – Q.10 Carry TWO marks Each

Passage Comprehension

Q.6	Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?
(A)	All human beings can digest dairy milk.
(B)	No human being can digest dairy milk.
(C)	Digestion of dairy milk is essential for human beings.
(D)	In human beings, digestion of dairy milk resulted from a mutated gene.
Q.7	The probability of a boy or a girl being born is 1/2. For a family having only three children, what is the probability of having two girls and one boy?
(A)	3/8
(B)	1/8
(C)	1/4
(D)	1/2

Probability

Sol^{n} :- 1 2 3 ← Three children

Case 1 :- G G B → 1/2 × 1/2 × 1/2 = 1/8

Case 2 :- G B G → 1/8

Case 3 :- B G G → 1/8

Total : 1/8 + 1/8 + 1/8

= 3/8 Ans

Quantitative Aptitude

Q.8 Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund A	Mutual fund B	Mutual fund C
Person 1	₹10,000	₹20,000	₹20,000
Person 2	₹20,000	₹15,000	₹15,000

At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

(A) 7.5%

(B) 10%

(C) 15%

(D) 20%

Soln

Person 1 - Person 2 = 500

Person 1 $10K(1+x) + 20K(1.15) + 20K(1.15)$

Person 2 $-\ 20K(1+x) + 15K(1.15) + 15K(1.15)$

Removed K by dividing by 1000

$$10K(1.15) - 10K - 10Kx = \frac{500}{1000} = 0.5$$

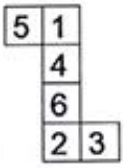
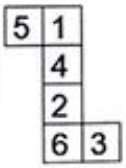
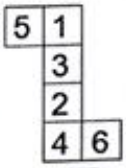
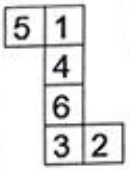
$$11.5 - 10 - 10x = 0.5$$

$$\Rightarrow 10x = 1$$

$$\Rightarrow x = 0.1 \quad (10\%)$$

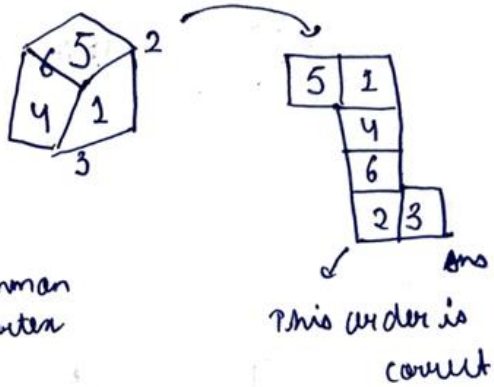
Ans

General Aptitude → Visual Reasoning

Q.9	Three different views of a dice are shown in the figure below.  The piece of paper that can be folded to make this dice is
(A)	
(B)	
(C)	
(D)	

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Soln Try to visualize a dice ⟩ It has 6 faces



← adjacent

5 & 6 shares a common vertex

Ans

This order is correct

Visual Reasoning

Q.10	Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?
(A) ✓	A rhombus
(B)	A triangle
(C)	An ellipse
(D)	A hexagon

Soln:- Visualise two birthday cone





→ Cross section is a Rhombus

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Q.11 – Q.35 Carry ONE mark Each

Maths

Probability

Q.11	Ten cards in a pack are numbered as 1, 2, 3, . . .10. The probability of drawing a card with an even number or a number which is a multiple of 5 from the pack is _____.
(A)	4/10
~~(B)~~	6/10
(C)	2/10
(D)	3/10

Soln Let's solve this probability question step-by-step

The numbers on the cards are : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Step 1 : Identify the even numbers & multiples of 5.

- Even Numbers : 2, 4, 6, 8, 10
- Multiples of 5 : 5, 10

Step 2 : Combine both sets of numbers & avoid counting any number twice

- Even numbers or multiple of 5 : 2, 4, 6, 8, 10, 5
  (6 numbers)

Step 3 : Calculate the probability

Total Cards = 10 Number of cards meeting the criteria = 6

$$\boxed{6/10}$$ Ans

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	Chemistry
Q.12	Hardness in water is **NOT** caused by _____.
(A)	Ca^{2+}
~~(B)~~	Si^{2+} (Actually, Ea (correct Answer)
(C)	Mg^{2+}
~~(D)~~	CO_3^{2-} (Given by GATE)

Hardness in water is primarily caused by the presence of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. Let's examine the given options:

(A) **Ca^{2+}** - Causes hardness in water. (B) **Si^{2+}** - Silicon ions do not typically cause hardness in water. (C) **Mg^{2+}** - Causes hardness in water. (D) **CO_3^{2-}** - Carbonate ions contribute to temporary hardness in water, but they are not the primary cause.

Based on this, the correct answer is (B) **Si^{2+}**. Silicon ions do not cause hardness in water.

Temporary and Permanent Hardness:

- **Temporary Hardness:** This is caused by dissolved bicarbonate minerals (calcium bicarbonate and magnesium bicarbonate). It can be removed by boiling the water, which causes the bicarbonates to precipitate out as carbonates.
- **Permanent Hardness:** This is caused by other dissolved minerals such as calcium sulfate and magnesium sulfate. Boiling does not remove these minerals.

Given Options: (A) **Ca^{2+} (Calcium ions):** These ions are a primary contributor to both temporary and permanent hardness in water. When dissolved in water, calcium ions make the water "hard."

(B) **Si^{2+} (Silicon ions):** Silicon ions do not typically contribute to water hardness. Silicon is more commonly found in water as silicic acid or silicate minerals, which do not contribute to the hardness measured by conventional means.

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(C) **Mg^{2+} (Magnesium ions):** These ions are another significant contributor to water hardness, similar to calcium ions. Magnesium ions are found in hard water and contribute to both temporary and permanent hardness.

(D) **CO_3^{2-} (Carbonate ions):** Carbonate ions are involved in the process of temporary hardness. When combined with calcium or magnesium, they form calcium carbonate ($CaCO_3$) or magnesium carbonate ($MgCO_3$), which can precipitate out of the water upon boiling. However, carbonates themselves are not the primary cause of hardness; rather, it's the calcium and magnesium ions that are.

Sr. No	Temporary hardness	Permanent hardness
1	Type of hardness which can be removed by simple techniques such as boiling is known as temporary hardness	Type of hardness which cannot be removed by simple techniques such as boiling is known as Permanent hardness
2	It is due to bicarbonates of Ca,Mg and carbonates of Mg hence known as carbonate hardness	It is due to other salts hence known as carbonate hardness
3	It is due to $Ca(HCO_3)_2$, $Mg(HCO_3)_2$, $MgCO_3$	It is due to $CaCl_2$,$MgCl_2$,$CaSO_4$, $MgSO_4$,$Ca(NO_3)_2$, $Mg(NO_3)_2$,FeC_{l2},
		&$FeSO_4$
4	It is known as alkaline hardness	It is known as non alkaline hardness
5	It is removed by boiling not by chemicals	It is removed by chemicals lime and soda
6	$Ca(HCO_3)_2 \rightarrow CaCO_3+CO_2+2H_2O$ $Mg(HCO_3)_2 \rightarrow Mg(OH)_2+2CO_2$	$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3+Na_2SO_4$

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Environmental Science and Engineering (ES)

Chemistry Ligany → d-orbital

Q.13	The maximum coordination number of Sn^{4+} is _____.
(A)	4
(B)	8
(C)	6
(D)	2

Maximum number of empty orbitals available for bonding

valency + ligands

$4 + 2 = 6$

The maximum coordination number of Sn^{4+} can be **4, 6, or 7**. The coordination number of 6 is the most common.

Explanation

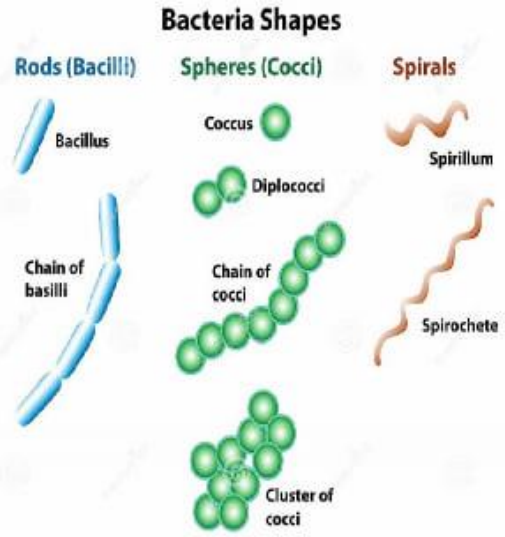
- **Coordination number**: The number of atoms, molecules, or ions that are bonded to a central atom in a molecule or crystal.
- Sn^{4+}: Sn^{4+} can occur in three coordination numbers, 4, 6, and 7.
- **Coordination number 6**: The coordination number 6 is the most common for Sn^{4+}.

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	Microbiology
Q.14	Rod shaped bacterial cells are called _____.
(A)	*Bacilli*
(B)	*Cocci*
(C)	*Spirilla*
(D)	*Diplococci* → 2 circular

Based on shape

- Cocci - spherical shaped (singular – coccus)
- Diplococci- pairs of cocci
- Streptococci- chains of cocci
- Staphylococci- bunches of cocci
- Bacilli- rod-shaped (singular – bacillus)
- Coccobacilli- with a shape that is intermediate between cocci and bacilli.
- Streptobacilli- a chain of bacilli
- Spirillum – spiral or corkscrew-shaped
- Vibrio - comma shaped
- Spirochete – wavy shaped



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Q.15	Tuberculosis is predominantly caused by _____.
(A)	*Entamoeba histolytica* → Amoebiasis
(B)	*Salmonella typhi* → Typhoid
(C)	*Mycobacterium bovis* → Tuberculosis
(D)	*Bacillus cereus* → Diarrhoea

Disease	Causative Microbe	Type
Tuberculosis (TB)	Mycobacterium tuberculosis	Bacterium
Cholera	Vibrio cholerae	Bacterium
Staphylococcus Infections	Staphylococcus aureus	Bacterium
Salmonellosis	Salmonella spp.	Bacterium
Syphilis	Treponema pallidum	Bacterium
Leprosy	Mycobacterium leprae	Bacterium
Gonorrhea	Neisseria gonorrhoeae	Bacterium
Tetanus	Clostridium tetani	Bacterium
Influenza (Flu)	Influenza virus	Virus
HIV/AIDS	Human Immunodeficiency Virus (HIV)	Virus
Hepatitis	Hepatitis viruses (A, B, C, D, E)	Virus
COVID-19	Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)	Virus
Ebola	Ebola virus	Virus
Measles	Measles virus	Virus
Zika Virus	Zika virus	Virus
Candidiasis	Candida spp.	Fungus
Aspergillosis	Aspergillus spp.	Fungus
Histoplasmosis	Histoplasma capsulatum	Fungus
Malaria	Plasmodium spp.	Parasite

	Env. Nitrogen Cycle
Q. 16	Which one of the following conversions belongs to nonsymbiotic nitrogen fixation?
(A)	Atmospheric nitrogen to ammonia by *Rhizobium* bacteria in nodules attached to roots of legumes Symbiotic Bacteria
(B)	Atmospheric nitrogen to ammonia by *Azotobacter* species Non Symbiotic
(C)	Nitrate to gaseous nitrogen under anaerobic conditions Denitrification
(D) X	Nitrate to ammonia under aerobic conditions
	Ammonia → (Nitrosomonas) Nitrite (NO_2^-) → (Nitrobacter) Nitrate

Nonsymbiotic nitrogen fixation is the process of converting atmospheric nitrogen (N_2) into ammonia (NH_3) or related compounds by free-living bacteria that do not form symbiotic relationships with plants.

Let's review the options:

(A) Atmospheric nitrogen to ammonia by Rhizobium bacteria in nodules attached to roots of legumes - This is symbiotic nitrogen fixation, as Rhizobium bacteria form a mutualistic relationship with legume plants.

(B) Atmospheric nitrogen to ammonia by Azotobacter species - This is an example of nonsymbiotic nitrogen fixation. Azotobacter species are free-living nitrogen-fixing bacteria.

(C) Nitrate to gaseous nitrogen under anaerobic conditions - This describes the process of denitrification, not nitrogen fixation.

(D) Nitrate to ammonia under aerobic conditions - This describes the process of ammonification or nitrate reduction, not nitrogen fixation.

Based on these explanations, the correct answer is:

(B) Atmospheric nitrogen to ammonia by Azotobacter species

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Water Engg. Repeated Q from Civil 2016 Q. paper

Q.17	Crown corrosion of reinforced cement concrete sewer is caused by _____.
(A)	sulfur oxidizing bacteria
(B)	iron oxidizing bacteria
(C)	denitrifying bacteria
(D)	fermentative bacteria

The crown corrosion in a reinforced concrete sewer is caused by the formation of H_2S gas due to aerobic and anaerobic conditions prevalent in the lower layers of wastewater, which contains sulphates, nitrogen, calcium, chloride etc. The reaction of sulphates involve in crown corrosion is given below.

$$SO_4^{2-} + \text{Organic matter} \xrightarrow[\text{Attack}]{\text{Bacterial}} S^{2-} + H_2O + CO_2$$

$$S^{2-} + 2H^+ \rightarrow H_2S$$

The H_2S gas produced is inflammable. This reacts with oxygen to form sulphuric acid.

$$H_2S + 2O_2 \xrightarrow[\text{Attack}]{\text{Bacterial}} H_2SO_4$$

H_2SO_4 is collected in the crown of sewer in a droplet form. It acts with concrete to form $CaSO_4$, which results in crown corrosion of reinforced concrete.

Hence, the correct option is (A).

Air

Q.18 The processes of removal of particles in a rapid sand filter with their description is given in the table.

Process	Description
(i) Straining	P: Removes only particles in the water large enough to get caught in the pores of the filter
(ii) Sedimentation	Q: Larger and heavier particles do not follow the fluid streamline around the sand grain and settle on the grain
(iii) Interception	R: Particles that do follow the streamline, but are too large and are caught because they brush up against the sand grains
(iv) Diffusion	S: Very small particles are experiencing Brownian motion and may collide with the sand grains by chance

Select the correct match.

(A) i- S; ii-P; iii-Q; iv-R

(B) i-Q; ii-R; iii-S; iv-P

(C) i-R; ii- S; iii- P; iv-Q

(D) i-P; ii-Q; iii-R; iv-S

Let's match the processes of particle removal in a rapid sand filter with their descriptions:

1. **Straining**:
   - Removes only particles in the water large enough to get caught in the pores of the filter.
2. **Sedimentation**:
   - Larger and heavier particles do not follow the fluid streamline around the sand grain and settle on the grain.
3. **Interception**:
   - Particles that do follow the streamline, but are too large and are caught because they brush up against the sand grains.
4. **Diffusion**:
   - Very small particles are experiencing Brownian motion and may collide with the sand grains by chance.

Now we can match the descriptions to the processes:

i. Straining → P ii. Sedimentation → Q iii. Interception → R iv. Diffusion → S

(D) **i-P; ii-Q; iii-R; iv-S**

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Air

Q.19	The environmental temperature increases by 6 °C/km with height at a particular location. The stability condition of the atmosphere at the location is _____.
(A)	stable
(B)	unstable
(C)	inversion
(D)	neutral

Decrease → Super ALR [unstable]
Sub ALR [Stable]
Neutral

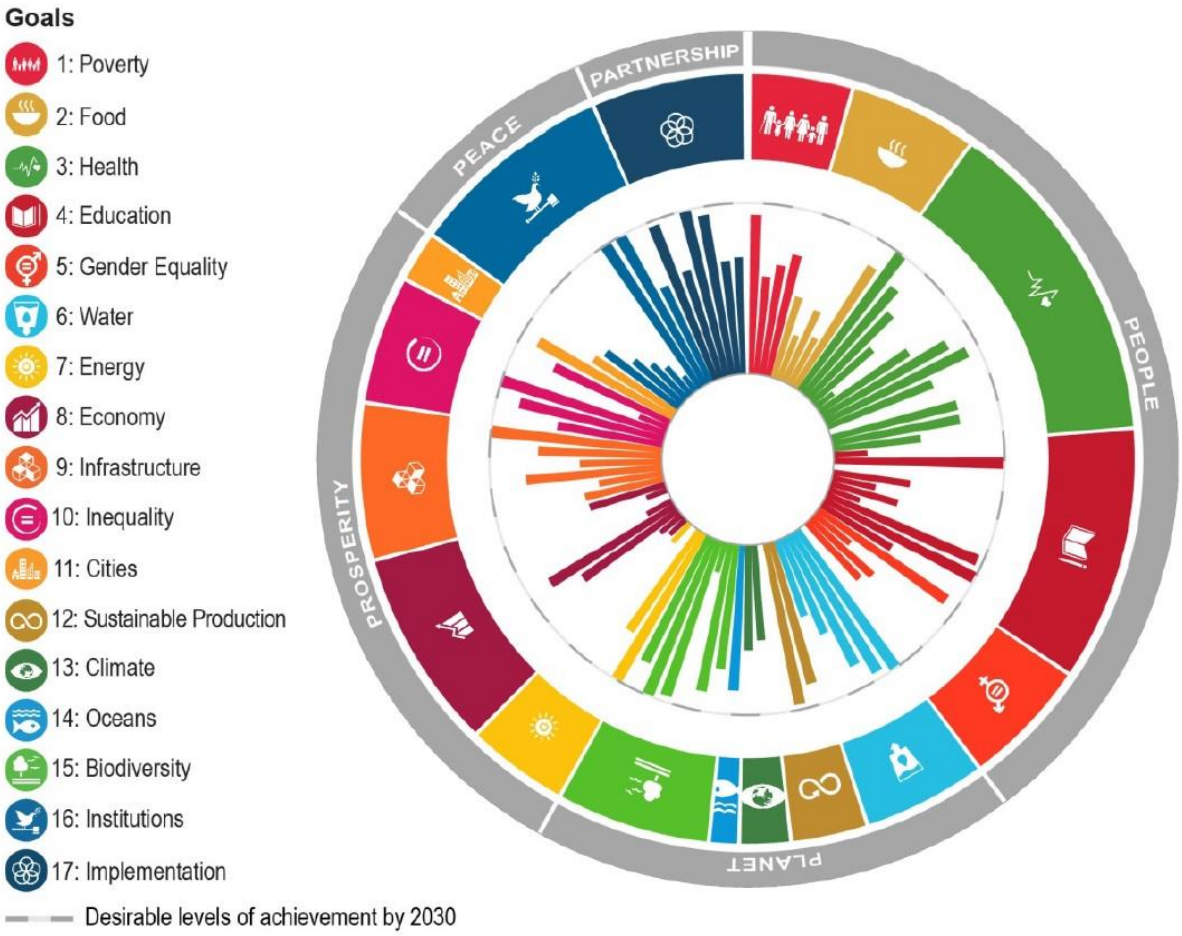
Increase → Inversion

Stability Condition	Description	Example	Effects
Stable Atmosphere	Displaced air returns to its original position.	Ball in a bowl	Prevents vertical mixing, promotes clear skies
Unstable Atmosphere	Displaced air continues to move away from its original position.	Ball on a hilltop	Enhances vertical mixing, promotes cloud formation and precipitation
Neutral Atmosphere	Displaced air stays at its new position without returning or moving further away.	Ball on a flat surface	Neutral effects on vertical motion and weather patterns
Temperature Inversion	Environmental temperature increases with height, creating a warm layer above cooler air near the surface.	-	Traps pollutants, reduces visibility, suppresses cloud formation and precipitation

	EMSD
Q.20	As per the United Nations agenda for sustainable development adopted in September 2015, the number of Sustainable Development Goals (SDGs) are _____ and the proposed target year to achieve them is _____.
(A)	15; 2035
(B)	17; 2030
(C)	20; 2050
(D)	18; 2047



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Environmental Science and Engineering (ES)

Air

Q.21	Which one of the following is **NOT** a greenhouse gas?
(A)	CO_2
(B)	CH_4
(C)	H_2S
(D)	H_2O

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	SWM
Q.22	As per the United Nations Environmental Program (UNEP) guidelines 2004, the maximum size of microplastics is _____.
(A)	10 mm
(B)	5 mm
(C)	10 µm
(D)	5 µm
Soln	Micro : 1 - 5 mm ← maximum Nano : < 1 µm

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Environmental Science and Engineering (ES)

SWM

Q.23	The costliest functional element in an urban centralized Municipal Solid Waste management infrastructure for a typical Indian Tier I city is _____.
(A)	biological treatment
(B)	collection and transport
(C)	disposal in a sanitary landfill
(D)	thermal treatment

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	Maths
Q.24	The eigen values of the matrix $\begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$ are (A, C)
(A)	1
(B)	2
(C)	7
(D)	4

$\begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$ Find eigen values

Sol^n: Its characteristic Eq: $|A - \lambda I| = 0$

$$\begin{vmatrix} 4-\lambda & 3 \\ 3 & 4-\lambda \end{vmatrix}$$

$(4-\lambda)^2 - 9 = 0$ $\quad (a-b)^2 = a^2 + b^2 - 2ab$

$16 + \lambda^2 - 8\lambda - 9 = 0$

$\lambda^2 - 8\lambda + 7 = 0$

$(\lambda - 7)(\lambda - 1)$ $\quad$ Eigen values:- 1, 7

$\lambda = 7, \lambda = 1$

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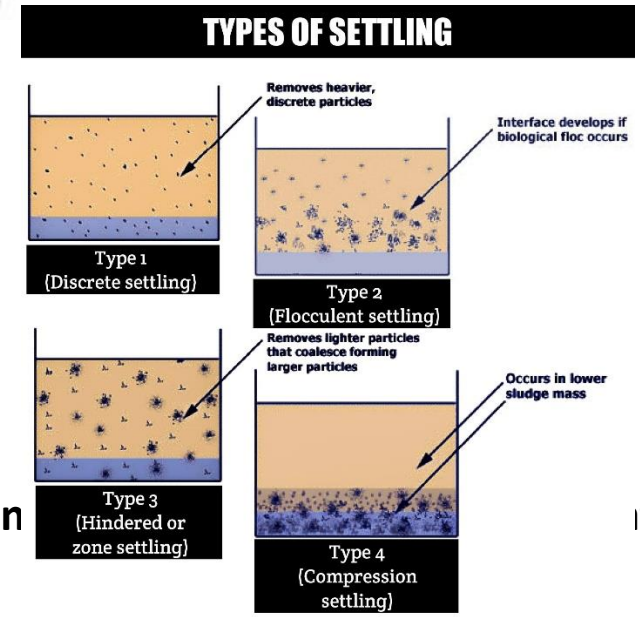
Maths

Q. 25	If **X** is a vector, and **A** and **B** are linear operators; then the correct mathematical relationship(s) is/are (A, B, C)
~~(A)~~	$(\mathbf{A}+\mathbf{B})\mathbf{X} = \mathbf{AX} + \mathbf{BX}$ → Linear operators distribute over addition. This property is linearity
~~(B)~~	$(\lambda\mathbf{A})\mathbf{X} = \lambda(\mathbf{AX})$ → Scalar multiplication distributes over linear operations. Here, λ is a scalar
~~(C)~~	$(\mathbf{AB})\mathbf{X} = \mathbf{A}(\mathbf{BX})$ → Associative property of linear operators
~~(D)~~	$(\mathbf{A}+\mathbf{B})\mathbf{X} = \mathbf{A}^T\mathbf{X}+\mathbf{B}^T\mathbf{X}$ Transpose are not needed & it alters the operations
	FM
Q.26	In the context of fluid flow, which of the following statement(s) is/are correct?
~~(A)~~	*Streamline* is a line, tangent to which at any point gives the direction of the velocity vector
(B)	*Streakline* → Pathline is the actual path traversed by a given fluid particle in an unsteady flow
~~(C)~~	*Streakline* and *streamline* are same for a steady flow
~~(D)~~	*Pathline* and *streamline* are same for a steady flow
	(A, C, D)

GATE 2024

OCF

Q.27	In a rectangular open channel, the flow is critical, and the flow depth is 2 m. Select the correct statement(s) (A, C)
(A)	Specific energy for the flow is 3.0 m — For rectangular channel $S.E = 1.5\, y_c = 3$
(B)	Specific energy for the flow is 2.0 m
(C)	Froude number is 1.0 — Sub critical $Fr < 1$; Critical $Fr = 1$; Super critical $Fr > 1$
(D)	Froude number is 1.5
	Wastewater Engg.
Q.28	With respect to particle settling in wastewater treatment systems; the correct statement(s) is/are (C, D)
(A)	Settling in grit chamber and primary sedimentation tanks are examples of Type-I settling — Type - I; Type - II
(B)	Settling in primary sedimentation tank and secondary sedimentation tank are examples of Type-II settling — Type - II; Type - III
(C)	Settling in grit chamber is an example of Type-I settling, whereas settling in primary sedimentation tank is an example of Type-II settling
(D)	Settling in secondary sedimentation tank is an example of Type-III settling, whereas settling in primary sedimentation tank is an example of Type-II settling



Telegran ıtion

Air

Q.29	The equipment that can be used to control particulate air pollution in an industrial unit is/are (solid) (A, B, C)
(A)	Electrostatic precipitator
(B)	Cyclone separator
(C)	Gravity settler
(D)	Incinerator → use in S.W.M for reducing moisture
	Air
Q.30	Which is/are the secondary air pollutant(s)? (A B D)
(A)	O_3
(B)	HNO_3
(C)	CO_2
(D)	H_2SO_4

GATE 2024

SWM

Q.31	As per the Hazardous Waste (Management and Handling) Rules, 2016, of India, which is/are the characteristic(s) that must be exhibited by a waste to be classified as a "characteristic" hazardous waste? (A, B, D) ✓
(A)	Ignitability
(B)	Reactivity
(C)	Radioactivity
(D)	Toxicity
	Maths
Q.32	$f(x) = x^3 - 4.5x^2 - 12x$ has a local maximum at $x =$ **−1** *(an integer value)* in the range $x = -2$ to $+2$.

1. Find the first derivation

$$f(x) = \frac{d}{dx}(x^3 - 4.5x^2 - 12x) = 3x^2 - 9x - 12$$

2. Set the first derivative equal to zero to find critical points

$$3x^2 - 9x - 12 = 0 \qquad \text{(Divide by 3)}$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$x = 4$ & $x = -1$ [As, 4 is outside the range of -2 to $+2$] are only consider = (−1)

Lets Check & Evaluate

$f(-1) = (-1)^3 - 4.5(-1)^2 - 12(-1) =$ **6.5** ← Maximum

$f(-2) = (-2)^3 - 4.5(-2)^2 - 12(-2) = -2$

$f(2) = (2)^3 - 4.5(2)^2 - 12(2) = -34$

The local maxima of the function $f(x)$ in the range $x = -2$ to $x = 2$ is at $x = -1$ Ans

Q.33	Consider the equation $\frac{dy}{dx} - x^2 + e^x = 0$; with $y = 1$ at $x = 0$. The value of y at $x = 1$ is 0.37 (*rounded off to 2 decimal places*). Take the value of e (base of natural logarithm) as 2.7.

Soln

$$\frac{dy}{dx} = x^2 - e^x \quad \text{(standard form)}$$

Integrate

$$\int dy = \int (x^2 - e^x)\,dx$$

$$y = \frac{(x)^3}{3} - e^x + C$$

$$y(1) = \frac{(1)^3}{3} - e^{\wedge 1} + 2$$

$$= 1/3 - 2.7 + 2$$

$$= 0.3667$$

$$\boxed{\sim 0.37}$$ Ans

Q.34	A municipal solid waste digester generates 1000 kg of methane gas. The volume of the tank needed to store this gas at 30 °C and 3 atmospheric pressure is 517625 liters (*an integer value*). Use R=0.082 L-atm/mole-K, Atomic weights of C=12, and H=1

Soln

$$Vol = 0.082 \times \frac{T}{P}$$

$$= 0.082 \times \frac{273+30}{3}$$

$$= 8.282\ L/mol$$

1 mole of CH_4 $\longrightarrow$ 8.282 L/mol

1 mole of CH_4 $\longrightarrow$ 16 g (MW)

? $\longrightarrow$ 1000×10^3 g

$$\frac{1000 \times 10^3}{16} = 62500 \text{ moles}$$

Volume of CH_4 occupied $= 62500 \times 8.282\ L$

$$= 517625\ L$$

Hydrology

~~Q.55~~	A Class-A pan was setup adjacent to a lake for measuring evaporation losses in the lake. The depth of water in the pan at the beginning of a certain week was 250 mm. In that week, there was a rainfall event with 10 mm depth. Water depth in the pan at the end of the week was 240 mm. The pan coefficient is 0.8. The estimated lake evaporation during the week was 16 mm (*an integer value*).

Soln

$S_1 = 250$ mm

$S_2 = 240$ mm

$P = 10$ mm

$P_{Co} = 0.8$

$\downarrow$

$Z = 10$ mm

difference between water level

(case II)

$E_p = P + Z = 10 + 10 = 20$ mm

$E_a = 20 \times 0.8$

$= 16$ mm

GATE 2024

Q.36 – Q.65 Carry TWO marks Each

Maths

Q.36

A population (with mean μ) follows normal distribution. Ten samples (N) are drawn at random with a mean value of "x" and standard deviation of "S". Following table provides the confidence limits, C(t) of the cumulative probability function for Student's t – distribution two-tailed test with degree of freedom, D.

Which one of the following expression is correct for testing the null hypothesis H_0: $\mu = 0$ at 10% significance level?

Cumulative probability

Degree of freedom

D	C(t)		
	0.9	0.95	0.975
9	1.38	1.83	2.26
10	1.37	1.81	2.23
11	1.36	1.80	2.20

(A) $-1.81 < \dfrac{x}{\dfrac{S}{\sqrt{N-1}}} < 1.81$

(B) $-1.83 < \dfrac{x}{\dfrac{S}{\sqrt{N-1}}} < 1.83$

(C) $-1.37 < \dfrac{x}{\dfrac{S}{\sqrt{N-1}}} < 1.37$

(D) $-2.23 < \dfrac{x}{\dfrac{S}{\sqrt{N-1}}} < 2.23$

Soln

$N = 10$. Degree of freedom $(N-1) = 9$.

10% significance level for two-tailed test

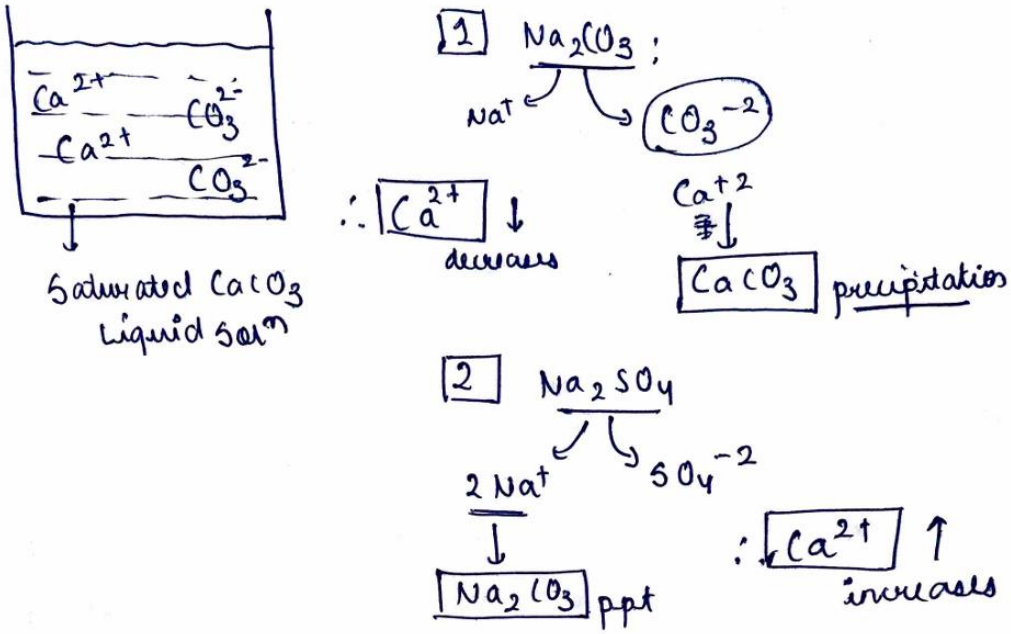
↳ 5% on left tail & 5% on right tail

∴ Cumulative probability = 0.95

& Degree of freedom = 9

Answer = 1.83

	Chemistry
Q.38	A saturated $CaCO_3$ stock solution is existing at 25°C. In one experiment (i) 25 g Na_2CO_3 is added to the stock solution. In another experiment (ii) 25 g Na_2SO_4 is added to the stock solution. Select the correct statement from the following.
(A)	Addition of (i) increases the concentration of Ca^{2+} and addition of (ii) decreases the concentration of Ca^{2+}
(B)	Addition of (i) decreases the concentration of Ca^{2+} and addition of (ii) increases the concentration of Ca^{2+}
(C)	Addition of (i) and (ii) increase the concentration of Ca^{2+}
(D)	Addition of (i) and (ii) decrease the concentration of Ca^{2+}



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	Kinetics & reactor design
Q.39	Consider second order kinetics $(r_c = -kC^2)$ under steady state condition. The ratio of volume of a complete mixed reactor (CMR) to that of a plug flow reactor (PFR) to achieve 90% reduction in the concentration is _____. ↳ Remaining → 10 %. Inlet concentrations in both the reactors are same.
(A)	10.0
(B)	1.0
(C)	0.1
(D)	2.3

Soln: 2nd order:

$$V_{CMR} = \frac{C_0}{K}$$

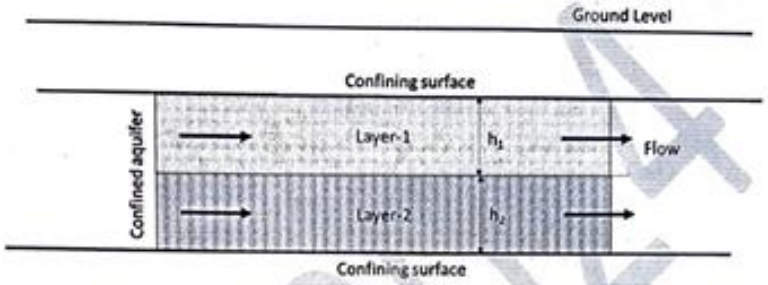
$$V_{PFR} = 1/K \left[1/C - 1/C_0\right] \qquad \boxed{C = 0.1\, C_0}$$

$$\frac{V_{CMR}}{V_{PFR}} = \frac{C_0 \cdot K}{K\left(1/0.1C_0 - \frac{1}{C_0}\right)} \simeq (10)$$

GATE 2024

Hydrology

Q.10 Consider two horizontal layers of an aquifer as shown in figure. Each layer is isotropic and homogeneous. Flow is parallel to the stratification. Thickness and horizontal hydraulic conductivity of layer-1 are h_1 and K_1, respectively. Thickness and horizontal hydraulic conductivity of layer-2 are h_2 and K_2, respectively, where h_1 is not equal to h_2. The equivalent horizontal conductivity K_x for the aquifer system is given by _____.



(A) $K_x = \dfrac{K_1.h_1 + K_2.h_2}{h_1 + h_2}$

(B) $K_x = \dfrac{K_1 + K_2}{2}$

(C) $K_x = \dfrac{K_1.h_2 + K_2.h_1}{h_1 + h_2}$

(D) $K_x = \sqrt{K_1.K_2}$

Soln To find the equivalent horizontal hydraulic Conductivity K_x for an aquifer system with two horizontal layers where the flow is parallel to the stratification, we use the weighted average formula.

Layer 1 : Thickness = h_1 & Hydraulic con. = K_1
Layer 2 : Hydraulic con. = K_2 & Thickness = h_2

$$\boxed{K_x = \frac{K_1.h_1 + K_2.h_2}{h_1 + h_2}}$$ Ans

	Air
Q.41	A gravity settling chamber of height 'H' and length 'L' is designed to control particulate air pollution. In the chamber, the horizontal velocity of air flow is 'V_h' and terminal settling velocity of the target particle is 'V_t'. Which one of the following expressions is the correct concept used to calculate the minimum size of the target particle that will be removed with 100% efficiency?
(A)	$\frac{V_t}{L} = \frac{V_h}{H}$
(B)	$V_h \times V_t = L \times H$
(C)	$V_h = V_t \times L \times H$
(D)	$\frac{V_t}{H} = \frac{V_h}{L}$

Soln

Key Parameters

H : Height of chamber
L : Length of the chamber
V_h : Horizontal velocity of air flow
V_t = Terminal settling velocity of the particle

Formula:

- Time to travel horizontally across the chamber

$$t_{horizontal} = \frac{L}{V_h}$$

- Time to settle vertically to the bottom of the chamber

$$t_{vertical} = \frac{H}{V_t}$$

For 100 efficiency

$$t_{vertical} = t_{horizontal}$$

$$\frac{H}{V_t} = \frac{L}{V_h}$$ can also be written as $$\boxed{\frac{V_t}{H} = \frac{V_h}{L}}$$ Ans

Q.43 Enzymes with the class of enzymes are listed in the table.

Enzyme	Class of Enzyme
(a) Lactate dehydrogenase	(i) Isomerases
(b) Alanine racemase	(ii) Transferases
(c) Lipase	(iii) Oxidoreductases
(d) Hexokinase	(iv) Hydrolases

Select the correct match(es)

(A) (a) - (iii); (b) - (i)

(B) (c) - (iv); (d) - (ii)

(C) (a) - (ii); (b) - (iv)

(D) (c) - (iii); (d) - (i)

Let's correctly match the given enzymes with their respective classes:

Enzyme Classes:

1. **Lactate dehydrogenase** - This enzyme belongs to the **oxidoreductases** class, which catalyzes oxidation-reduction reactions.
    - Match: (a) - (iii)
2. **Alanine racemase** - This enzyme belongs to the **isomerases** class, which catalyzes the rearrangement of atoms within a molecule.
    - Match: (b) - (i)
3. **Lipase** - This enzyme belongs to the **hydrolases** class, which catalyzes the hydrolysis of various bonds.
    - Match: (c) - (iv)
4. **Hexokinase** - This enzyme belongs to the **transferases** class, which catalyzes the transfer of functional groups from one molecule to another.
    - Match: (d) - (ii)

Correct Matches:

- (a) Lactate dehydrogenase - (iii) Oxidoreductases
- (b) Alanine racemase - (i) Isomerases
- (c) Lipase - (iv) Hydrolases
- (d) Hexokinase - (ii) Transferases

So, the correct answers are:

(A) (a) - (iii); (b) - (i) (B) (c) - (iv); (d) - (ii)

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	Micro
Q.44	With reference to disinfection, which of the following statement(s) is/are **CORRECT**?
(A)	Ethanol damages lipid structures in the bacterial cell membrane.
(B)	Mercuric chloride inactivates cellular enzymes containing sulfhydryl groups.
(C)	Glutaraldehyde inactivates protein.
(D)	Isopropyl alcohol cannot be used as a disinfectant.

A, B, C

Solution

- (A) Ethanol damages lipid structures in the bacterial cell membrane.
- (B) Mercuric chloride inactivates cellular enzymes containing sulfhydryl groups.
- (C) Glutaraldehyde inactivates protein.
- (D) Isopropyl alcohol cannot be used as a disinfectant.

The correct answers are marked as A, B, and C. The incorrect answer is D, as isopropyl alcohol can indeed be used as a disinfectant.

Q.45 Which of the following statement(s) is/are **CORRECT**? A, C

(A)	DNA is composed of nucleotides.
(B)	Five types of nitrogenous bases occur in DNA.
(C)	Each phosphate is attached to two deoxyribose units in a single strand of DNA.
(D)	The ratio of adenine to guanine is always 1:1 in a double stranded DNA.

let's go through each option one by one:

1. **(A) DNA is composed of nucleotides:**
   - This statement is correct. DNA, or deoxyribonucleic acid, is made up of molecules called nucleotides. Each nucleotide contains a phosphate group, a sugar molecule (deoxyribose), and a nitrogenous base. The four types of nitrogenous bases in DNA are adenine (A), thymine (T), cytosine (C), and guanine (G). These nucleotides are arranged in two long strands that form a double helix.
2. **(B) Five types of nitrogenous bases occur in DNA:**
   - This statement is incorrect. DNA contains four types of nitrogenous bases: adenine (A), thymine (T), cytosine (C), and guanine (G). While RNA (ribonucleic acid) contains uracil (U) instead of thymine, DNA does not have uracil, and there are no additional nitrogenous bases in DNA beyond these four.
3. **(C) Each phosphate is attached to two deoxyribose units in a single strand of DNA:**
   - This statement is correct. In the DNA backbone, each phosphate group connects two deoxyribose (sugar) molecules. This connection is formed through phosphodiester bonds, which link the 3' carbon atom of one sugar molecule to the 5' carbon atom of the next sugar

molecule. This creates a repeating pattern of sugar-phosphate-sugar-phosphate along the DNA strand.

4. **(D) The ratio of adenine to guanine is always 1:1 in double stranded DNA:**
    - This statement is incorrect. The ratio of adenine to guanine is not 1:1. In double-stranded DNA, the base pairing rules are that adenine (A) pairs with thymine (T) and cytosine (C) pairs with guanine (G). Therefore, the ratio of adenine to thymine is 1:1, and the ratio of cytosine to guanine is also 1:1. However, there is no fixed ratio between adenine and guanine.

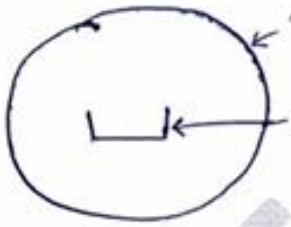
So, the correct statements about DNA are (A) and (C).

	Wastewater Engg.
Q.46	The Streeter–Phelp's oxygen sag equation for a river is based on a few assumptions. The correct assumption(s) is/are
(A)	At any instant the deoxygenation rate is directly proportional to the amount of oxidizable organic material present.
(B)	At any instant the deoxygenation rate is inversely proportional to the amount of oxidizable organic material present.
(C)	The reoxygenation rate is directly proportional to the dissolved oxygen deficit.
(D)	The reoxygenation rate and deoxygenation rate are directly proportional to the saturation concentration of dissolved oxygen. ↳ No correlation

OCF

~~Q.47~~	Water is flowing **FULL** through a rectangular tunnel of size 3 m (width) × 2 m (height). The average velocity of flow is 1 m/s. The frictional head loss is observed to be 1 m per km. Consider acceleration due to gravity (g) as 10 m/s^2. The correct statement(s) is/are
~~(A)~~	Hydraulic radius is 0.6 m
~~(B)~~	Darcy-Weisbach friction factor is 0.048
~~(C)~~	Hydraulic radius is 2 m
~~(D)~~	Darcy-Weisbach friction factor is 0.024

Soln



$$R = \frac{By}{B+2y}$$

$$R = \frac{3 \times 2}{3+4} = 0.857 = R$$

$$h_f = \frac{f l v^2}{2gy}$$ (Darcy velocity)

y in place of D

$$f = \frac{2gyh_f}{Lv^2}$$

$$= \frac{2 \times 10 \times 2 \times 1}{1000 \times 1^2} = 0.04 = f$$

Darcy-Weisbach friction factor (f)

GATE 2024

EMSD

Q.48 Based on the ISO 14040 methodology for Life Cycle Assessment, match the terms with the descriptions in the table.

A, D

Term	Description
(a) Goal and Scope	(i) Based on the product or system, the comparative unit must be carefully defined and be same for all scenarios
(b) Functional Unit	(ii) The problem is described, and the objective of the study are defined
(c) Life Cycle Inventory	(iii) Evaluates the environmental implications due to the inventorized emissions
(d) Impact Assessment	(iv) Process based approach and input-output approach

The correct match(es) is/are

(A) (a)-(ii); b-(i);

(B) (a)-(iii), b-(i)

(C) (c)-(iii), (d)-(iv)

(D) (c)-(iv), (d)-(iii)

GATE 2024

	Chemistry
Q.50	The pH of a solution containing 0.1M of acetic acid and 0.05 M of sodium acetate is 4.46 (*rounded off to 2 decimal places*). The pK_a value of ionization of acetic acid is 4.76.

Q.50

Soln Acid = [0.1] ← Acetic acid

Salt = [0.05] ← sodium acetate

pKa = 4.76

$$pH = pKa + \log \frac{[Salt]}{[Acid]}$$

$$= 4.76 + \log \frac{0.05}{0.1}$$

$$= 4.76 - 0.30$$

pH = 4.46

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	Chemistry
Q.51	The ionic strength of a solution containing 0.01M of $CaCl_2$ and 0.001M of Na_2SO_4 is 0.033 M (*rounded off to 3 decimal places*).

Soln

$$I = 1/2 \sum m_i z_i^2$$

where:

m_i is the molarity of ion i

z_i is the charge of ion i

1. $CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$

Molarity of $Ca^{2+} = 0.01M$

Molarity of $Cl^- = 2 \times 0.01 = 0.02M$

Charge of $Ca^{2+} = 2$

Charge of $2Cl^- = 1$

2. $Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$

Molarity of $2Na^+ = 2 \times 0.01 = 0.02M$

Molarity of $SO_4^{2-} = 0.01M$

Charge of $2Na^+ = 1$

Charge of $SO_4^{2-} = 2$

Substitute into formula:

$$I = 1/2 \left(0.01 \times (2)^2 + 0.02 \times (1)^2 + 0.002 \times (1)^2 + 0.001 \times (2^2)\right)$$

$$I = 1/2 (0.066)$$

$$\boxed{I = 0.033}$$ Ans

	Chemistry / Air
Q.52	The concentration of Ozone corresponding to a mixing ratio of 120 ppbv at pressure of 1 atmosphere and temperature of 25°C is 235.71 µg/m³ (*rounded off to 1 decimal place*). Atomic weight of oxygen = 16; R= 8.314 J/K-g.mole.

(52) Soln

$O_3 \rightarrow 3 \times 16 = 48$ mw

$V = 0.082 \times \frac{T}{P}$

$V = 24.436$

O_3 (µg/mol) $= 120 \times \frac{48}{24.436}$

$O_3 = 235.71$ µg/m³

Q.53 One million liters per day (MLD) of wastewater with a soluble BOD of 200 mg/L is treated in an activated sludge process. The BOD of treated wastewater is 20 mg/L. The observed yield coefficient of the biological system is (0.35) Y_t/Y_u

The daily biomass generation in the system is 63 kg (*an integer value*).

(53) Soln

Total Mass = 1 MLD × (200 − 20) mg/L = 180 kg/D

Biomass in the system = 180 × 0.35 = 63 kg/D Ans

Q.54 An industry discharges 2 million liters per day (MLD) of wastewater with a temperature of 45°C and a pH of 2, whereas the neighboring industry produces 3 MLD of wastewater with a temperature of 30°C and pH of 8. If both the wastewaters are mixed and carried through a pipeline, then the resultant pH of mixed wastewater is 2.39 (*rounded off to 2 decimal places*).

H^+ → 10^{-2}, 10^{-8}

Neglect buffering capacity of the system and the temperature effect on pH.

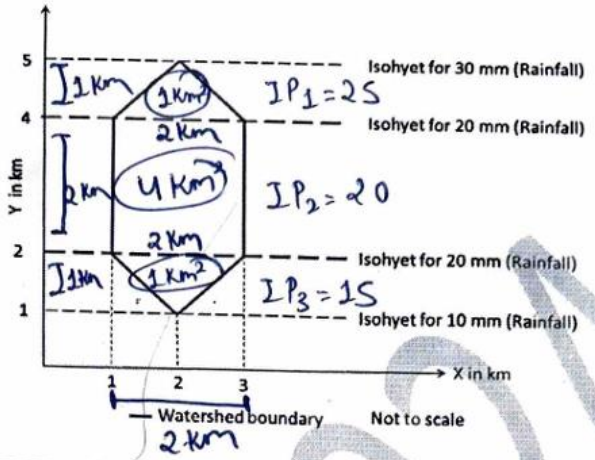
Mix pH $= -\log_{10}\left[\frac{Q_1 H_1 + Q_2 H_2}{Q_1 + Q_2}\right]$

$= -\log_{10}\left[\frac{2 \times 10^{-2} + 3 \times 10^{-8}}{2 + 3}\right]$

$= 2.39$ Ans

Hydrology

Q.55 Consider a watershed and isohyets as shown in the figure. The average rainfall in the watershed is 20 mm (an *integer value*).



$$P_{AVG} = \frac{(25 \times 1) + (20 \times 4) + (15 \times 1)}{6}$$

$$= 20 \text{ mm}$$

GATE 2024 IISc Bengaluru

Environmental Science and Engineering (ES)

Air

Q.57 In a cyclone separator of radius 25 cm, a particle is travelling with a gas stream at velocity of 18 m/s. The ratio of centrifugal force to the gravitational force acting on the particle is 132.24 (*rounded off to 2 decimal places*).

Consider acceleration due to gravity (g) as 9.8 m/s².

Soln

Centrifugal force -

$$F_c = m \cdot \frac{v^2}{r}$$

Gravitational force

$$F_g = m \cdot g$$

Ratio of centrifugal force to gravitational force

$$\frac{F_c}{F_g} = \frac{m \cdot v^2/r}{m \cdot g} \qquad \frac{F_c}{F_g} = \frac{v^2}{r \cdot g}$$

$$\frac{F_c}{F_g} = \frac{18^2}{0.25 \times 9.8}$$

$$= \frac{324}{2.45}$$

$$= \boxed{132.24} \text{ Ans}$$

Noise

Q.58 Two sources of noise, adjacent to each other in a room, have sound pressure levels of 30 and 40 decibel (dB). The combined sound pressure level in the room is 40 dB (*rounded off to 2 decimal places*).

Use reference sound pressure as 20 μPa.

30 dB
40 dB
→ 10 different = change 0
dB = 40

Air

Q.59 An industrial stack emits 100 g/s of CO at an effective height of 'H', where the wind speed is 5 m/s. At 3 km distance downwind, the values of dispersion coefficient in y-direction and z-direction are 50 m and 25 m, respectively. The CO concentration at the centerline of the plume at 3 km distance downwind is 2.52 mg/m³ (*rounded off to 2 decimal places*)?

NO @ CL

Use Gaussian plume model and value of π = 3.14. Neglect reactions and the ground effect of plume in the calculations.

Soln

$$C = \left[\frac{Q}{2\pi\sigma_y\sigma_z H}\right]\left[e^{-1/2(y/\sigma_y)^2}\right]\left[e^{-1/2(H/\sigma_z)^2}\right]$$

$$C = \left[\frac{100}{2\times3.14\times50\times25\times5}\right]\left[e^{-1/2(3/50)^2}\right]\left[e^{-1/2(3/25)^2}\right]$$

$$C = 2.52\times10^{-3}\ g/m^3$$

$$\boxed{C = 2.52\ mg/m^3}$$ Ans

↳ units given in question

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	SWM
Q.60	Two hypothetical organic waste streams A and B are mixed prior to the composting process. Waste-A has 2.16% of C and 1.20% of N. Waste-B has 19.10% of C and 0.14% of N. The quantity of Waste-B that should be mixed with per kg of Waste-A to achieve the desired C:N ratio of 25 is 1.785kg (*rounded off to 2 decimal places*). Assume both the waste streams are completely dry.

Sol^{n}

1. Calculate the amount of carbon & N in waste - A & waste - B:

For waste - A (1 kg):

Carbon : 2.16% of 1 kg = 0.0216 kg

Nitrogen : 1.20% of 1 kg = 0.012 kg

For waste - B (let the quantity be x kg):

Carbon : 19.10% of x kg = 0.191x kg

Nitrogen : 0.14% of x kg = 0.0014x kg

2. Make a equation

Desired C:N ratio = 25

$$\frac{C}{N} \quad \frac{0.0216 + 0.191x}{0.012 + 0.0014x} = 25$$

Find the value of x

$$0.0216 + 0.191x = 25\,(0.012 + 0.0014x)$$

$$0.0216 + 0.191x = 0.3 + 0.035x$$

$$0.191x - 0.035x = 0.3 - 0.0216$$

$$0.156x = 0.2784$$

$$x = \frac{0.2784}{0.156}$$

$$\boxed{x \simeq 1.785}$$ Ans

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SWM

Q.61	Food waste, paper waste and plastic waste have typical densities of 280 kg/m^3, 80 kg/m^3, and 50 kg/m^3, respectively. The mixed waste is composed of 70% food waste, 20% paper waste and 10% plastic waste. The density of the mixed waste is 142.85 kg/m^3 (*rounded off to 2 decimal places*). Neglect compaction effect.

Soln

$$\rho_{mix} = \frac{(280 \times 70) + (80 \times 20) + (50 \times 10)}{100}$$

$$= \boxed{142.85 \text{ kg/m}^3} \quad \text{Ans}$$

Water Engg.

Q.63	A person consumes 2.5 liters of water per day. The water quality test indicated that the supplied water has a Pb concentration of 0.6 mg/L. If the weight of the person is 75 kg, the exposure level for Pb for this person from this drinking water source is 0.02 mg/kg/day (*rounded off to 2 decimal places*).

= Body weight

Soln

$$\text{Exposure level (mg/kg/day)} = \frac{\text{Amount of Pb consumed per day (mg)}}{\text{Body weight (kg)}}$$

$$\text{Pb consumed per day (mg)} = 0.6 \text{ mg/L} \times 2.5 \text{ L} = 1.5 \text{ mg/day}$$

Body weight = 75 kgs

$$\text{Exposure level} = \frac{1.5 \text{ mg/day}}{75 \text{ kg}} = \boxed{0.02 \text{ mg/kg/day}} \quad \text{Ans}$$

Irrigation

Q.64 In a region, total annual consumption of gasoline is 30.6 million tons. The land required for growing sugarcane to produce enough bioethanol to replace the gasoline completely is 143,300 km^2 *(an integer value)*.

Ethanol energy equivalent is 67% of gasoline, gasoline density is 850 kg/m^3, yield of bioethanol produced from sugarcane per hectare of land is 3750 L, and 1 km^2 = 100 hectares.

Soln

1. Convert Gasoline Consumption to Liters:

$$\text{Volume} = \frac{30.6 \times 10^6 \times 10^3 \text{ kg}}{850 \text{ kg/m}^3} \times 1000 \text{ L/m}^3 = 36 \times 10^9 \text{ L}$$

(million, tons)

2. Determine the Amount of Ethanol Needed:

Ethanol = 67% of gasoline

$$\text{Volume of ethanol needed} = \frac{36 \times 10^9 \text{ L}}{0.67} \approx 53.73 \times 10^9 \text{ L}$$

3. Calculate the Land Required for Bioethanol Production:

Yield of bioethanol from sugarcane per hectare = 3750 L

1 km^2 = 100 hectares

$$\text{Land (in hectares)} = \frac{53.73 \times 10^9 \text{ L}}{3750 \text{ L/hectare}} \approx 14.33 \times 10^6 \text{ hectares}$$

$$\text{Land (in km}^2\text{)} = \frac{14.33 \times 10^6 \text{ hectares}}{100} \approx \boxed{143,300 \text{ km}^2}$$ Ans

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For Verification, You can check the official Answer Key

GRADUATE APTITUDE TEST IN ENGINEERING 2024
अभियांत्रिकी स्नातक अभिक्षमता परीक्षा २०२४
ORGANISING INSTITUTE: INDIAN INSTITUTE OF SCIENCE, BENGALURU

Environmental Science & Engineering (ES)
Final Answer Key

Q. No.	Session	Question Type	Section	Key/Range	Mark
1	1	MCQ	GA	D	1
2	1	MCQ	GA	A	1
3	1	MCQ	GA	B	1
4	1	MCQ	GA	B	1
5	1	MCQ	GA	B	1
6	1	MCQ	GA	D	2
7	1	MCQ	GA	A	2
8	1	MCQ	GA	B	2
9	1	MCQ	GA	A	2
10	1	MCQ	GA	A	2
11	1	MCQ	ES	B	1
12	1	MCQ	ES	D	1
13	1	MCQ	ES	C	1
14	1	MCQ	ES	A	1
15	1	MCQ	ES	C	1
16	1	MCQ	ES	B	1
17	1	MCQ	ES	A	1
18	1	MCQ	ES	D	1
19	1	MCQ	ES	C	1
20	1	MCQ	ES	B	1
21	1	MCQ	ES	C	1
22	1	MCQ	ES	B	1
23	1	MCQ	ES	B	1
24	1	MSQ	ES	A;C	1
25	1	MSQ	ES	A;B;C	1
26	1	MSQ	ES	A;C;D	1
27	1	MSQ	ES	A;C	1
28	1	MSQ	ES	C;D	1
29	1	MSQ	ES	A;B;C	1
30	1	MSQ	ES	A;B;D	1
31	1	MSQ	ES	A;B;D	1
32	1	NAT	ES	-1 to -1	1

33	1	NAT	ES	-0.40 to -0.30	1
34	1	NAT	ES	517625 to 517885	1
35	1	NAT	ES	16 to 16	1
36	1	MCQ	ES	B	2
37	1	MCQ	ES	D	2
38	1	MCQ	ES	B	2
39	1	MCQ	ES	A	2
40	1	MCQ	ES	A	2
41	1	MCQ	ES	D	2
42	1	MSQ	ES	B;C	2
43	1	MSQ	ES	A;B	2
44	1	MSQ	ES	A;B;C	2
45	1	MSQ	ES	A;C	2
46	1	MSQ	ES	A;C	2
47	1	MSQ	ES	;B	2
48	1	MSQ	ES	A;D	2
49	1	NAT	ES	3.70 to 3.90	2
50	1	NAT	ES	4.46 to 4.46	2
51	1	NAT	ES	0.033 to 0.033	2
52	1	NAT	ES	234.0 to 236.0	2
53	1	NAT	ES	63 to 63	2
54	1	NAT	ES	2.38 to 2.41	2
55	1	NAT	ES	20 to 20	2
56	1	NAT	ES	3.40 to 3.50	2
57	1	NAT	ES	132.00 to 133.00	2
58	1	NAT	ES	40.00 to 41.00	2
59	1	NAT	ES	2.50 to 2.60	2
60	1	NAT	ES	1.75 to 1.80	2
61	1	NAT	ES	142.00 to 143.00	2
62	1	NAT	ES	320.00 to 321.00	2
63	1	NAT	ES	0.02 to 0.02	2
64	1	NAT	ES	143280 to 144000	2
65	1	NAT	ES	155 to 155	2

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